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**A PROJECT BASED LEARNING
REPORT ON**

“MineLab”

Submitted by

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For the course Data Warehouse and Mining

Submitted To Course Coordinator

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CERTIFICATE

This is to certify that the Project Based Learning work entitled “**MineLab**” has been completed by Harshit Bhardwaj under the guidance of Prof Rohini Khalkar for the course Data Warehouse and Mining

The project presents an interactive simulation of page replacement policies and demonstrates the organization and management of memory resources. The work is submitted towards the partial fulfilment of the academic requirements of the Bachelor of Technology degree in Computer Engineering at Bharati Vidyapeeth (Deemed to be University), College of Engineering, Pune

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I extend my sincere appreciation to the faculty members who provided their guidance and support throughout the development of the project. Their instruction in **data preprocessing, data quality analysis, association rule mining, classification, regression, and clustering**, along with their emphasis on understanding algorithmic principles, preprocessing requirements, and evaluation measures, contributed significantly to the design and implementation of **MineLab**.

ABSTRACT

MineLab is an integrated **Data Warehousing and Data Mining analytics workspace** developed as a practical mini project for the DWM laboratory. It brings major DWM concepts into a unified workflow, allowing users to load **CSV, XLSX, or XLS datasets**, or use built-in samples. The system profiles and validates data, identifies attribute types and roles, detects quality issues, applies preprocessing, and provides dedicated modules for **classification, regression, clustering, and association rule mining**.

The preprocessing module supports **missing-value handling, duplicate removal, categorical encoding, numerical scaling, outlier treatment, and train-test splitting**. It also provides attribute detection for numerical, categorical, boolean, date, and identifier fields, along with statistics such as missing values, unique values, and numerical summaries.

For classification, MineLab implements **J48 Decision Tree, Naive Bayes, and Logistic Regression**, reporting accuracy, precision, recall, F1-score, confusion matrices, and per-class results. The regression module provides **Simple and Multiple Linear Regression** with MAE, MSE, RMSE, R^2 , predictions, regression equations, and visualizations. **K-Means clustering** supports configurable cluster settings and evaluates results using **Silhouette Score and WCSS**, along with centroids and cluster summaries.

The association-mining module implements **Apriori and FP-Growth** for frequent-itemset and rule discovery. Generated rules are evaluated using **support, confidence, lift, and conviction**. The **Results Hub** provides centralized comparison of algorithms across their relevant evaluation metrics, supported by visualizations such as confusion matrices, regression plots, clustering graphs, correlation heatmaps, feature-importance charts, and association-rule networks. Results can also be exported in **CSV, Excel, PNG, and PDF** formats.

The project includes synthetic datasets for demonstrating different DWM tasks, including a **50,000-row customer churn dataset** for classification and preprocessing analysis, along with datasets for regression, clustering, association mining, and preprocessing. Overall, MineLab demonstrates the complete DWM workflow—from **dataset loading and quality analysis to preprocessing, mining, evaluation, visualization, and report generation**—within a single interactive academic application.

1. INTRODUCTION AND PROJECT OVERVIEW

1.1 Background

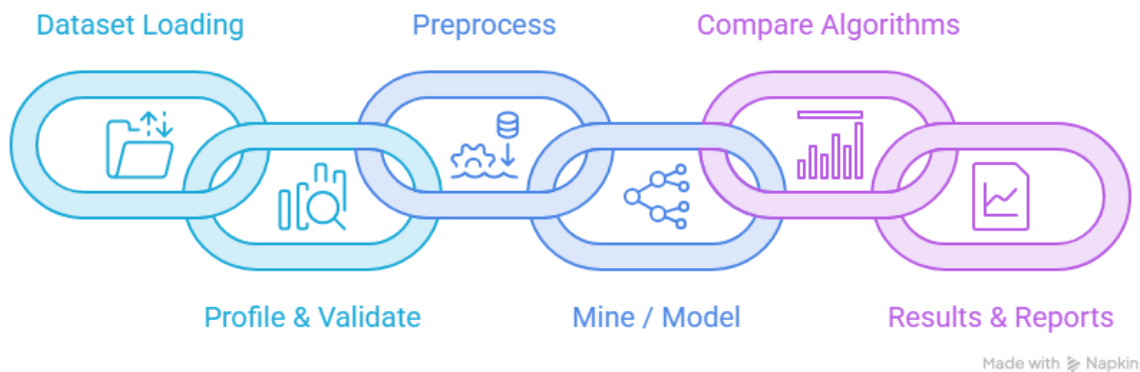
Modern organizations generate large volumes of data through business transactions, applications, databases, sensors, online platforms and other digital systems. This data may contain numerical values, categorical attributes, identifiers, dates and other descriptive information. When such data is stored only as raw rows in spreadsheets or CSV files, it can be difficult to identify meaningful patterns, understand data quality issues, make predictions or discover relationships between attributes.

Data Warehousing focuses on collecting, organizing and preparing data so that it can be used effectively for analysis, while Data Mining focuses on extracting useful patterns, relationships and predictive knowledge from that data. MineLab was developed with the idea that these concepts should be demonstrated as parts of a connected analytical workflow rather than as isolated laboratory exercises. The application allows a dataset to move through a common pipeline covering data loading, validation, attribute detection, preprocessing, mining, evaluation, visualization and reporting.

The project provides an interactive environment for studying multiple Data Mining techniques through a unified interface. The system supports classification using J48 Decision Tree, Naive Bayes and Logistic Regression; regression using Simple Linear Regression and Multiple Linear Regression; clustering using K-Means; and association rule mining using Apriori and FP-Growth. Each technique is accompanied by appropriate evaluation metrics and visualizations so that users can understand not only the final result but also the working and performance of the algorithm.

MineLab is designed as an academic analytics workspace rather than a production enterprise system. It includes built-in synthetic datasets for demonstrating preprocessing, classification, regression, clustering and association analysis, while also allowing users to upload their own CSV and Excel datasets. This makes the application reusable for different laboratory experiments and educational scenarios while maintaining a consistent workflow from raw data to analytical interpretation.

MineLab Analytical Workflow



1.2 Project Vision and Key Contribution

The central vision of MineLab is to make the complete Data Warehousing and Data Mining workflow visible through a single interactive academic environment. A user should not need to use separate applications for dataset preparation, classification, regression, clustering and association rule mining. Instead, MineLab maintains a common active dataset and provides dedicated modules for each stage of the analytical process. The dashboard provides an overall view of the project, Dataset Overview and Attribute Detection explain the structure and quality of the data, Data Preprocessing prepares the working dataset, the mining modules perform analytical tasks, and the Results Hub, Visualization and Reporting features convert the results into understandable analytical outputs.

The architecture of MineLab follows a modular client-side design. The main application manages navigation, application state and interaction, while separate core modules handle dataset loading, attribute detection, statistics, preprocessing, validation and evaluation metrics. Individual algorithm modules implement J48 Decision Tree, Naive Bayes, Logistic Regression, Simple Linear Regression, Multiple Linear Regression, K-Means, Apriori and FP-Growth. This separation allows each analytical component to be developed and tested independently while still operating through the same application workflow.

The integrated Results Hub is an important feature of the project because it allows the results of multiple mining techniques to be viewed and compared from a common interface.

Classification algorithms can be evaluated using accuracy, precision, recall and F1-score, regression algorithms using MAE, MSE, RMSE and R^2 , clustering using silhouette score and WCSS, and association rule mining using rule count, support, confidence, lift and conviction.

2. LITERATURE REVIEW AND TECHNOLOGY CONTEXT

2.1 Data Warehousing and Analytical Processing

Data Warehousing refers to the systematic collection, integration and organization of data for analytical processing and decision support. A data warehouse generally brings together data from different sources and provides a structured environment in which historical data can be analyzed. Concepts such as data integration, data quality, transformation, multidimensional analysis and analytical reporting form an important foundation for Data Warehousing.

In a typical data warehousing environment, analytical data may be organized using structures such as fact tables and dimension tables. Fact tables generally contain measurable business events, while dimension tables provide descriptive information used to analyze those events across different perspectives. Although MineLab does not implement a persistent enterprise-level data warehouse, it demonstrates the analytical side of the DWM workflow by allowing datasets to be loaded, profiled, cleaned, transformed and analyzed within a unified environment.

The project therefore focuses on the practical relationship between data preparation and analytical processing. Dataset validation, attribute detection, statistical analysis and preprocessing form the foundation for the subsequent data mining operations. This provides an academic demonstration of how properly prepared data can be transformed into useful analytical information.

2.2 Data Mining

Data Mining is the process of discovering useful patterns, relationships, trends and predictive structures from datasets. A typical data mining workflow includes data cleaning, transformation, feature preparation, model construction, evaluation and interpretation. Different mining techniques are selected according to the type of analytical problem being addressed.

MineLab incorporates four major categories of data mining tasks. Classification is used to assign records to predefined categories using J48 Decision Tree, Naive Bayes and Logistic Regression. Regression is used to predict numerical target values using Simple Linear Regression and Multiple Linear Regression. Clustering is an unsupervised learning technique implemented through K-Means to identify naturally occurring groups within numerical data. Association Rule Mining is used to discover relationships between items using Apriori and FP-Growth.

The project also emphasizes the importance of preprocessing before applying mining algorithms. Missing-value handling, duplicate removal, categorical encoding, numerical scaling and outlier treatment can significantly affect the quality of analytical results. By allowing different preprocessing configurations to be applied before mining, MineLab provides a practical way to observe the relationship between data quality, preprocessing and algorithm performance.

2.3 Technology Context

MineLab is implemented as a browser-based client-side application using standard web technologies. HTML provides the structural foundation of the application, CSS is used for layout and interface styling, and JavaScript manages application logic, interaction, data processing and communication between the different modules. The application follows a modular design in which core data-processing utilities, mining algorithms, visualization functions, reporting and export functionality are separated into dedicated modules.

The project uses Papa Parse for reading and processing CSV files and SheetJS (XLSX) for handling Excel-based datasets. These libraries allow users to import common tabular data formats directly into the application. The application also uses Chart.js for generating interactive charts and analytical visualizations. These visualizations include dataset statistics, confusion matrices, feature-importance charts, regression plots, clustering visualizations, correlation heatmaps and association-rule networks.

The data mining algorithms are implemented within the application's JavaScript modules rather than being exposed through a separate backend service. The core modules provide functionality for dataset loading, attribute detection, statistics, preprocessing, validation and evaluation metrics, while dedicated algorithm modules implement J48 Decision Tree, Naive Bayes, Logistic Regression, Simple Linear Regression, Multiple Linear Regression, K-Means, Apriori and FP-Growth.

The application also uses html2pdf.js for generating PDF reports and browser-based export functionality for CSV, Excel and image outputs. Application settings and selected history information can be maintained using browser localStorage. This technology stack allows MineLab to operate as a self-contained academic data mining workspace without requiring a separate server-side machine-learning environment.

The modular architecture also makes the system suitable for laboratory demonstrations because each stage of the DWM workflow can be observed independently while remaining connected

to the same active dataset. This provides a practical implementation of the concepts studied in Data Warehousing and Data Mining using an accessible web-based environment.

Table 1: Project Technology Stack

Layer	Technology	Role
Frontend	HTML + CSS + JavaScript	Application structure, interface, navigation and user interaction
Data Import	Papa Parse + SheetJS	CSV and Excel dataset loading and processing
Data Processing	Custom JavaScript Modules	Data validation, attribute detection, statistics and preprocessing
Data Mining	Custom JavaScript Algorithms	J48, Naive Bayes, Logistic Regression, Linear Regression, Multiple Linear Regression, K-Means, Apriori and FP-Growth
Mining and Tooling	Custom Apriori + Vite + npm	Association mining and integrated development workflow
Reporting & Export	html2pdf.js + Export Utilities	PDF reports, CSV/XLSX exports and chart/image exports
Storage	Browser localStorage	Application settings and selected history data

3. PROBLEM DEFINITION, OBJECTIVES AND SCOPE

3.1 Problem Definition

The problem addressed by MineLab is the fragmentation of common Data Warehousing and Data Mining experiments in an academic environment. In a typical laboratory workflow, a student may use different tools for data cleaning and preprocessing, WEKA or another application for classification, separate programs or notebooks for regression and clustering, and additional tools for visualization and report preparation. Although each experiment demonstrates an individual concept, this fragmented approach makes it difficult to understand the relationship between data preparation, mining algorithms, evaluation metrics and final analytical results. Changing the dataset may also require repeating the workflow across multiple applications.

MineLab addresses this problem by integrating the major stages of the DWM workflow into a single interactive web-based application. The system accepts an active dataset, validates and profiles its structure, detects attribute types and possible roles, provides configurable preprocessing operations and then exposes dedicated modules for classification, regression, clustering and association rule mining. The objective is not to replace production-grade analytical platforms, but to provide an understandable, reusable and technically coherent academic implementation of the Data Warehousing and Data Mining concepts covered in the laboratory syllabus.

3.2 Objectives

The main objectives of the MineLab project are:

- To provide a unified web-based environment for demonstrating the complete Data Warehousing and Data Mining workflow.
- To support loading of datasets in CSV, XLSX and XLS formats along with built-in synthetic datasets for experimentation.
- To validate datasets and analyze their structure, data types, missing values, duplicates and basic statistical properties.
- To automatically detect numerical, categorical, boolean, date and identifier attributes and identify possible target attributes.

- To provide preprocessing operations including missing-value handling, duplicate removal, categorical encoding, numerical scaling and outlier treatment.
- To implement classification using J48 Decision Tree, Naive Bayes and Logistic Regression.
- To implement Simple Linear Regression and Multiple Linear Regression for numerical prediction.
- To implement K-Means clustering and evaluate the resulting clusters using silhouette score and WCSS.
- To discover frequent itemsets and association rules using Apriori and FP-Growth.
- To evaluate mining algorithms using task-specific performance metrics such as accuracy, precision, recall, F1-score, MAE, MSE, RMSE, R^2 , silhouette score, support, confidence, lift and conviction.
- To provide graphical visualizations for model performance, predictions, clusters, correlations and association rules.
- To provide a centralized Results Hub for comparing the outputs of different algorithms within their respective task categories.
- To support CSV, Excel, PNG and PDF export so that analytical results can be used for academic documentation and reporting.

3.3 Scope of the Project

The scope of MineLab covers the academic Data Mining lifecycle from dataset ingestion and data-quality analysis to preprocessing, model execution, evaluation, visualization and report generation. The input layer supports both built-in synthetic datasets and user-uploaded CSV, XLSX and XLS files. The dataset analysis layer calculates row and column counts, identifies numerical and categorical attributes, detects possible identifiers and target candidates, reports missing values and duplicates, and generates basic statistical summaries.

The preprocessing scope includes removal of duplicate records, handling of missing values through row removal or mean, median and mode imputation, categorical encoding through label encoding or one-hot encoding, numerical scaling through standardization or Min-Max

normalization, and optional outlier handling using IQR or Z-score based methods. Users can also configure the training and testing split and inspect the changes in dataset quality before and after preprocessing.

The mining scope covers four major analytical families. Classification includes J48 Decision Tree, Naive Bayes and Logistic Regression. Regression includes Simple Linear Regression and Multiple Linear Regression. Clustering includes K-Means with configurable cluster settings and evaluation through silhouette score and WCSS. Association rule mining includes Apriori and FP-Growth with rule evaluation using support, confidence, lift and conviction.

The visualization and reporting scope includes confusion matrices, feature-importance charts, regression plots, residual analysis, cluster visualizations, elbow curves, correlation heatmaps, association-rule networks, data-quality charts and algorithm comparison charts. The application also supports exporting analytical data and generated reports in commonly used academic formats.

The project is intended primarily as an academic and educational system. It does not attempt to provide production-grade authentication, enterprise-level data warehousing infrastructure, distributed data processing, real-time streaming analytics, continuous model training or large-scale cloud deployment. The application is designed to run interactively on a local environment and uses synthetic datasets where appropriate so that the complete workflow can be demonstrated without requiring confidential organizational data.

3.4 Target Users

The primary users of MineLab are students and teachers studying or demonstrating Data Warehousing and Data Mining concepts. Students can use the application to perform laboratory experiments, understand preprocessing and algorithm behavior, compare evaluation metrics and prepare practical examination demonstrations or reports. Teachers can use the system to demonstrate the relationship between an input dataset, data-quality issues, preprocessing operations, algorithm configuration and analytical results in a single environment.

A secondary target group is learners who want to experiment with their own educational datasets. By uploading a custom CSV or Excel dataset, users can observe how different preprocessing configurations and mining algorithms affect the resulting analysis.

4. DATA WAREHOUSING AND DATA MINING FOUNDATIONS

4.1 Data Warehouse Concepts

A data warehouse is designed to organize data from different sources in a structured form so that it can be analyzed efficiently. In a typical warehouse environment, data can be organized around business subjects such as sales, customers, products, finance or operations, with measurable facts connected to descriptive dimensions.

MineLab uses a dataset-oriented approach to demonstrate these concepts rather than implementing a permanent database warehouse. The application accepts structured datasets containing numerical, categorical, date and identifier attributes and analyzes their relationships through profiling, preprocessing, aggregation and visualization. This provides a practical representation of how organized analytical data can be transformed into information useful for decision-making.

The project also demonstrates the importance of separating measurable attributes from descriptive attributes. Numerical attributes can be used for statistical analysis, regression and clustering, while categorical and other descriptive attributes can be encoded and used by classification and other mining algorithms. The Attribute Detection module helps identify these roles automatically, making the relationship between data structure and analytical processing visible to the user.

4.2 ETL Thinking

The Extract, Transform and Load (ETL) process provides a useful conceptual framework for understanding the MineLab workflow.

Extraction occurs when the application reads a built-in dataset or imports a user-provided CSV, XLSX or XLS file. The Dataset Loader converts the input into a structured representation that can be processed by the application.

Transformation takes place during dataset validation, attribute detection and preprocessing. This stage can include missing-value handling, duplicate removal, categorical encoding, numerical scaling and outlier treatment. The system also generates statistics and data-quality information so that the effect of the transformations can be observed.

Loading in the MineLab prototype refers to making the validated and processed dataset available as the active dataset for subsequent analytical modules. The processed data can then be passed to classification, regression, clustering or association rule mining algorithms.

Although MineLab is designed as an academic application rather than a production data warehouse, this workflow demonstrates the fundamental ETL principle of converting raw source data into a structured and analysis-ready form.

4.3 Multidimensional Analysis and OLAP Concepts

Multidimensional analysis allows data to be examined from different perspectives by using dimensions, measures and aggregations. Common OLAP operations include **slice, dice, drill-down and roll-up**, which allow users to move between different levels of analytical detail.

MineLab demonstrates these concepts through its visualization and analytical modules. Users can select attributes for graphical analysis, group records, apply aggregation functions such as count, sum, mean, minimum, maximum and median, and examine relationships between different attributes. For example, a numerical attribute can be aggregated according to a categorical attribute to compare groups, while a date-related attribute can be used to examine trends over time.

A **slice** can be represented by focusing on a selected subset of the dataset or attribute. A **dice** operation can be demonstrated by applying multiple filtering conditions. **Drill-down** can be represented by moving from an aggregated view toward more detailed records or categories, while **roll-up** can be represented by aggregating detailed records into higher-level summaries.

The project's visualization module supports these analytical perspectives through charts such as bar charts, line charts, scatter plots, histograms, pie or donut charts and heatmaps. Therefore, although MineLab does not implement a dedicated OLAP server, it demonstrates the core analytical thinking behind multidimensional analysis and connects DWM theory with practical data exploration.

Overall, MineLab demonstrates that effective data mining depends not only on the algorithms themselves but also on how data is extracted, organized, transformed and presented. The project connects Data Warehousing concepts such as ETL, structured analytical data and multidimensional analysis with Data Mining operations such as classification, regression, clustering and association rule discovery.

4.4 Data Mining Task Types

Table 2: Data Mining Task Types Used in RetailSense

Task	Question	RetailSense Technique	Primary Output
Association Rule Mining	Which items or attributes frequently occur together?	Apriori, FP-Growth	Frequent itemsets, support, confidence, lift, conviction, association rules
Classification	Which class should a new record be assigned to?	J48 Decision Tree, Naive Bayes, Logistic Regression	Accuracy, precision, recall, F1-score, confusion matrix, predictions
Regression	What numerical value should be predicted?	Simple Linear Regression, Multiple Linear Regression	MAE, MSE, RMSE, R^2 , predictions, regression equation

The four major analytical task types supported by MineLab address different objectives and therefore require different evaluation approaches. Classification focuses on assigning records to predefined classes, regression predicts continuous numerical values, clustering discovers groups of similar records without predefined labels, and association rule mining identifies relationships between items or attributes.

4.5 Relationship Between Data Preparation and Mining

The quality of a data mining result depends significantly on the quality and representation of the input data. Missing values can reduce the reliability of statistical calculations and machine-learning models. Duplicate records can distort frequency-based analysis and may give excessive importance to repeated observations. Categorical attributes often need to be encoded before they can be processed by numerical machine-learning algorithms, while numerical attributes with significantly different scales can affect distance-based methods such as K-Means.

MineLab therefore treats preprocessing as an important stage of the overall mining workflow. The preprocessing module provides operations for missing-value handling, duplicate removal, categorical encoding, numerical scaling and outlier treatment. Users can select the required operations and observe the changes between the original and processed dataset before running the analytical algorithms.

The project also demonstrates an important distinction between data validation, preprocessing and modelling. Data validation and profiling identify the structure and quality of the available data, including data types, missing values, duplicates and statistical characteristics.

Preprocessing then transforms the working dataset according to the selected rules. Finally, the classification, regression, clustering or association modules operate on the prepared active dataset. This separation helps demonstrate why data preparation is an essential part of a reliable data mining workflow.

The workflow can therefore be viewed as a repeated analytical cycle:

Understand the dataset → Identify the analytical objective → Prepare the data → Select an appropriate algorithm → Run the analysis → Evaluate the results → Interpret the findings

This cycle is more important than memorizing individual algorithms because it demonstrates the reasoning required to select and apply a suitable data mining technique to a particular problem.

MineLab also includes synthetic datasets that can be used to demonstrate the effect of preprocessing. The datasets contain controlled characteristics such as missing values and different attribute types, allowing users to observe how preprocessing operations change data quality and structure. The larger 50,000-row customer churn dataset provides an additional benchmark for examining classification and preprocessing behavior on a more substantial dataset.

The project therefore emphasizes that data mining is not simply the process of executing an algorithm. Reliable analysis requires understanding the source data, identifying quality issues, selecting appropriate transformations, choosing a suitable mining technique and interpreting the resulting metrics in context.

5. SYSTEM ANALYSIS AND REQUIREMENTS

5.1 Functional Requirements

MineLab shall allow the user to load a built-in synthetic dataset or upload a dataset in CSV, XLSX or XLS format. The system shall validate the uploaded data, create an active dataset and display its basic structure, including row count, column count, attribute types, missing values, duplicate records and statistical information. The user shall also be able to reset the working dataset and begin analysis with another dataset.

The system shall automatically detect and classify attributes based on their characteristics. It shall identify numerical, categorical, boolean, date, string and identifier attributes and suggest possible target attributes where appropriate. The system shall provide dataset previews and data-quality summaries before mining operations are performed.

The system shall provide preprocessing controls for missing-value handling, duplicate removal, categorical encoding, numerical scaling and outlier treatment. Missing values shall support row removal and mean, median or mode imputation. Categorical attributes shall support label encoding and one-hot encoding. Numerical attributes shall support standardization and Min-Max normalization, while outlier processing shall support IQR and Z-score based methods. The user shall also be able to configure the training and testing split where applicable.

The system shall provide separate analytical modules for classification, regression, clustering and association rule mining. Classification shall support J48 Decision Tree, Naive Bayes and Logistic Regression. Regression shall support Simple Linear Regression and Multiple Linear Regression. Clustering shall support K-Means, while association rule mining shall support Apriori and FP-Growth.

Each analytical module shall provide task-specific configuration options and evaluation measures. Classification shall report accuracy, precision, recall, F1-score and confusion matrices. Regression shall report MAE, MSE, RMSE and R^2 . K-Means shall provide cluster assignments, cluster sizes, centroids, silhouette score and WCSS. Association rule mining shall provide frequent itemsets and rules evaluated using support, confidence, lift and conviction.

The system shall provide a centralized Results Hub where outputs from the implemented algorithms can be reviewed and compared. It shall also provide visualizations appropriate to each task, including confusion matrices, feature-importance charts, regression plots, residual

analysis, cluster visualizations, elbow curves, correlation heatmaps and association-rule networks.

The system shall support exporting analytical results and reports in appropriate formats including CSV, Excel, PNG and PDF. The generated report shall summarize the dataset, preprocessing configuration, algorithm configuration, results, comparisons and conclusions.

Table 3: System Functional Requirements

ID	Requirement	Expected Behaviour
FR1	Dataset Loading	Load built-in datasets or user-provided CSV, XLSX and XLS files.
FR2	Dataset Validation	Validate the dataset and identify issues such as empty data, invalid structure, duplicate headers and excessive missing values.
FR3	Dataset Profiling	Display row count, column count, data types, missing values, duplicates, unique values and basic statistics.
FR4	Attribute Detection	Identify numerical, categorical, boolean, date, string and identifier attributes and suggest possible target attributes.
FR5	Missing-Value Handling	Remove rows with missing values or apply mean, median or mode imputation according to the selected configuration.
FR6	Duplicate Handling	Detect and remove duplicate records when the corresponding preprocessing option is selected.
FR7	Encoding	Transform categorical attributes using label encoding or one-hot encoding.
FR8	Scaling	Apply standardization or Min-Max normalization to suitable numerical attributes.
FR9	Outlier Handling	Detect or treat outliers using IQR or Z-score based methods.
FR10	Classification	Execute J48 Decision Tree, Naive Bayes and Logistic Regression and report classification metrics.
FR11	Regression	Execute Simple Linear Regression and Multiple Linear Regression and report regression metrics.

5.2 Non-Functional Requirements

Usability:

MineLab is primarily intended for academic demonstrations, laboratory work and practical examinations. The interface should therefore remain simple, readable and consistent. Navigation between dataset analysis, preprocessing, mining, results, visualization and reporting should be straightforward. Important metrics and algorithm outputs should be presented in a form that can be easily interpreted during a demonstration or viva.

Performance:

The application should provide reasonable response times for the supplied synthetic datasets and typical educational datasets. The algorithms operate on data available within the application environment, making the system suitable for laboratory-scale experimentation. Execution time can also be displayed where applicable so that users can observe the computational cost of different configurations.

Portability:

The project should be capable of running as a self-contained web application in a modern browser. The use of browser-compatible JavaScript libraries for CSV/Excel processing, visualization and report generation reduces dependence on specialized external analytical software.

Maintainability:

The application is organized into separate modules for dataset loading, attribute detection, statistics, preprocessing, validation, metrics, algorithms, visualization, reporting, storage and export. This modular organization allows individual components or algorithms to be modified without requiring the complete application to be redesigned.

Reliability:

The system should validate datasets before allowing analytical operations to proceed and should provide meaningful messages when required attributes, targets or numerical features are unavailable. Preprocessing and model configurations should also be reflected in the generated results so that the analytical output can be reproduced within the application workflow.

6. DATASET AND DATA WAREHOUSE DESIGN

6.1 Demo Dataset Overview

MineLab includes a set of bundled demo datasets designed to demonstrate different stages and tasks of Data Warehousing and Data Mining. Instead of relying on a single retail transaction dataset, the system provides multiple datasets for exploring data preprocessing, classification, regression, clustering, and association rule mining.

The bundled datasets include the Preprocessing Impact Demo, Iris dataset, Market Basket dataset, and Housing Prices dataset. In addition, MineLab provides a 50,000-record Customer Churn dataset for demonstrating classification and the effect of data preprocessing on model performance.

The datasets are intended for academic and demonstration purposes. They provide varied data characteristics so that users can observe how different preprocessing techniques and mining algorithms behave on different types of data. This also allows MineLab to demonstrate the complete workflow from dataset loading and validation to preprocessing, mining, evaluation, visualization, and report generation within a single environment.

6.2 Data Quality Profile

MineLab is designed to work with datasets having different structures and data-quality characteristics. The bundled examples intentionally cover small educational datasets as well as a larger 50,000-row benchmark. This allows the user to observe how validation, attribute detection and preprocessing behave across different data sizes and attribute types.

The most important benchmark for the supervised-learning portion is the bundled customer dataset. It contains 50,000 rows and 19 columns, including 13 numerical and 6 categorical attributes. The selected classification target is Customer_Churn with the classes No and Yes, while Avg_Order_Value is used as the numerical regression target. The dataset contains 52,913 missing cells and no duplicate rows, making it suitable for demonstrating missing-value preprocessing without requiring duplicate removal in the representative benchmark run.

6.3 Conceptual Data Organization

MineLab does not implement a persistent enterprise data warehouse or a physical star schema. Instead, it uses a dataset-oriented in-memory analytical representation that demonstrates the practical relationship between data ingestion, transformation and mining. The same active dataset can be inspected, preprocessed and routed to several analytical task modules without being copied into separate applications.

For reporting purposes, the dataset can be viewed through three conceptual layers: source data, an active analysis representation, and task-specific analytical outputs. The source layer consists of built-in datasets or uploaded CSV/XLSX/XLS files. The active layer contains the validated and profiled dataset together with the selected preprocessing state. The analytical layer contains model outputs, metrics, visualizations and exported reports.

Table 6: Conceptual Analytical Layers

Layer	Role	MineLab Representation
Source Layer	Original data input	Built-in sample datasets or user-uploaded CSV/XLSX/XLS
Active Dataset Layer	Validated working data	Current rows, columns, detected attributes, statistics and preprocessing state
Mining Layer	Task-specific analysis	J48, Naive Bayes, Logistic Regression, Linear Regression, Multiple Linear Regression, K-Means, Apriori and FP-Growth
Presentation Layer	Interpretation and export	Results Hub, charts, tables, CSV/XLSX/PNG/PDF outputs

Table 7: Analytical Role of Dataset Attributes

Attribute Role	Examples	Use in MineLab
Identifier	customer_id / similar ID-like fields	Detected as identifier-like and normally excluded from ordinary modelling features
Numerical Feature	age, income, usage, charges	Statistical summaries, regression, clustering and model features
Categorical Feature	gender, country, contract_type	Encoding and categorical modelling
Boolean Feature	has_credit_card, is_active_member	Detected as boolean and used where suitable
Classification Target	Customer_Churn	Categorical output for J48, Naive Bayes and Logistic Regression

7. DATA PREPROCESSING AND QUALITY MANAGEMENT

7.1 Need for Preprocessing

Raw tabular data can contain missing values, duplicate records, categorical variables, different numerical scales and unusual observations. MineLab therefore treats preprocessing as a configurable stage rather than assuming that every dataset is already model-ready. The same preparation layer is reused by classification, regression and clustering, while association mining uses transaction-oriented preparation when required.

Table 8: Preprocessing Techniques

Technique	Purpose	MineLab Implementation
Missing-value handling	Reduce incomplete data	Remove rows or impute using mean, median or mode
Duplicate removal	Prevent repeated observations from distorting analysis	Detect and remove exact duplicate rows when enabled
Encoding	Convert categorical values to model-ready representations	Label Encoding or One-Hot Encoding
Scaling	Put numerical variables on comparable scales	Standardization (Z-score) or Min-Max normalization
Outlier handling	Identify or treat unusual numerical observations	IQR or Z-score based processing
Train / test split	Create held-out evaluation data	70/30, 75/25 or 80/20 according to configuration
Target protection	Avoid corrupting selected outputs	Selected classification/regression targets are protected from ordinary feature transformations

7.3 Duplicate Removal and Outlier Handling

The bundled 50,000-row customer benchmark contains zero exact duplicate rows. This does not make duplicate removal unnecessary as a MineLab feature: when a user uploads a dataset containing repeated records, the same preprocessing control can identify and remove them. The operation is therefore dataset-dependent and is not reported as a correction to the benchmark when no duplicates are present.

Outlier processing is optional and is implemented using IQR or Z-score based methods. For IQR-based detection, the system examines the first quartile (Q1), third quartile (Q3) and

interquartile range ($IQR = Q3 - Q1$). Values beyond the selected threshold can be identified for treatment. The purpose is to demonstrate the statistical method and its effect on the working dataset rather than to assume that every unusual observation is an error.

7.4 Before-and-After Demonstration

The representative preprocessing run on the 50,000-row customer benchmark uses median imputation, explicit missing-value indicators and one-hot encoding, while duplicate removal, scaling and outlier removal remain disabled for that run. The number of observations remains 50,000, while the feature representation expands from 19 source columns to 54 working columns. This is an important demonstration that preprocessing can alter model inputs even when the number of records does not change.

Table 9: Representative Preprocessing Outcome

Measure	Before	After	Change
Rows	50,000	50,000	No observation loss
Columns	19	54	Feature representation expanded
Missing cells	52,913	0 after imputation	Imputed in representative run
Duplicate records	0	0	No duplicates in bundled benchmark
Missing indicators	0	17	Indicator features added
One-hot encoded feature columns	0	26	Categorical representation expanded

Figure 5: Effect of preprocessing on dataset shape and feature representation

The processed dataset becomes the working representation for subsequent analytical operations. This establishes the pipeline dependency used throughout MineLab: the quality and representation of data presented to a mining algorithm depend on the preprocessing decisions selected by the user.

8. IMPLEMENTATION AND MODULE INTEGRATION

8.1 Frontend and Application Implementation

The application is implemented as a browser-based single-page workspace using HTML, CSS and JavaScript. `index.html` defines the application shell, `styles.css` provides the visual language, and `app.js` coordinates navigation, active state and the main user interactions. The project avoids a server-side machine-learning layer so that the complete analytical workflow can be demonstrated locally in the browser.

Table 15: Implementation Modules

Module Group	Representative Files	Responsibility
Application shell	<code>index.html</code> , <code>app.js</code> , <code>styles.css</code>	Page structure, navigation, state and user interaction
Data handling	<code>core/dataset-loader.js</code>	Import CSV/XLSX/XLS and maintain dataset representation
Detection and profiling	<code>core/attribute-detector.js</code> , <code>statistics.js</code> , <code>validation.js</code>	Types, roles, statistics, data-quality checks
Preprocessing	<code>core/preprocessing.js</code>	Missing values, duplicates, encoding, scaling, outliers and split
Algorithms	<code>algorithms/*.js</code>	J48, Naive Bayes, Logistic Regression, Simple/Multiple LR, K-Means, Apriori, FP-Growth
Metrics	<code>core/metrics.js</code>	Classification, regression and clustering evaluation
Visualization	<code>visualization/charts.js</code>	General charts and model-specific visualizations
Reporting / export	<code>reports/report-generator.js</code> , <code>utils/export.js</code>	PDF-ready reports and CSV/XLSX/PNG outputs
Local persistence	<code>utils/storage.js</code>	Browser localStorage for settings and selected history

8.2 Core and Algorithm Module Implementation

The core JavaScript modules separate reusable data operations from individual algorithms. The dataset loader converts imported files into a common row-oriented representation. Attribute

detection examines columns to distinguish numerical, categorical, boolean, date-like and identifier-style data. The statistics and validation modules calculate data-quality information before mining starts.

The preprocessing module then applies the user-selected transformations. Classification uses a dedicated matrix-preparation path that learns categorical encodings from the training portion and protects the selected target. Regression uses numerical target validation and builds matrices for one-feature or multiple-feature prediction. K-Means prepares numerical features and evaluates the resulting partitions, while Apriori and FP-Growth convert transaction-oriented input into item collections for frequent-pattern mining.

The reporting and visualization modules consume the results generated by these algorithms instead of retraining models independently for every display. This reduces unnecessary repeated computation and keeps displayed metrics tied to the analytical result returned by the corresponding module.

8.3 Integrated Application Execution

Because MineLab is client-side, execution begins by opening the web application in a modern browser. The user can immediately select a bundled dataset or upload CSV, XLSX or XLS data. There is no requirement for a separate API service or external database for the academic workflow. Browser-compatible libraries such as Papa Parse and SheetJS handle tabular import, Chart.js handles visualization, and html2pdf.js supports report generation.

This architecture makes the project suitable for classroom demonstrations and practical examinations because the major DWM concepts can be observed through one application. It also simplifies portability: the same source package can be used on a local machine without reproducing a multi-service deployment.

10.4 Handling Practical Data and Environment Issues

Several implementation issues were addressed through defensive data handling. Mixed numerical and categorical data is handled through attribute detection and explicit encoding. Missing data is represented using indicators and an explicit imputation policy. Identifier-like and high-cardinality fields are excluded from inappropriate modelling paths. Selected targets are protected so preprocessing intended for input features does not silently corrupt the output variable.

9. RESULTS, EVALUATION AND ALGORITHM COMPARISON

9.1 Dashboard and Results Hub Evidence

The MineLab Results Hub is designed to show classification and regression outputs together while still preserving the task-specific metric definitions. The supplied project screenshot demonstrates this interface state: three classification models appear on the left and two regression models appear on the right, with metric cards, charts and result tables available in the same view.

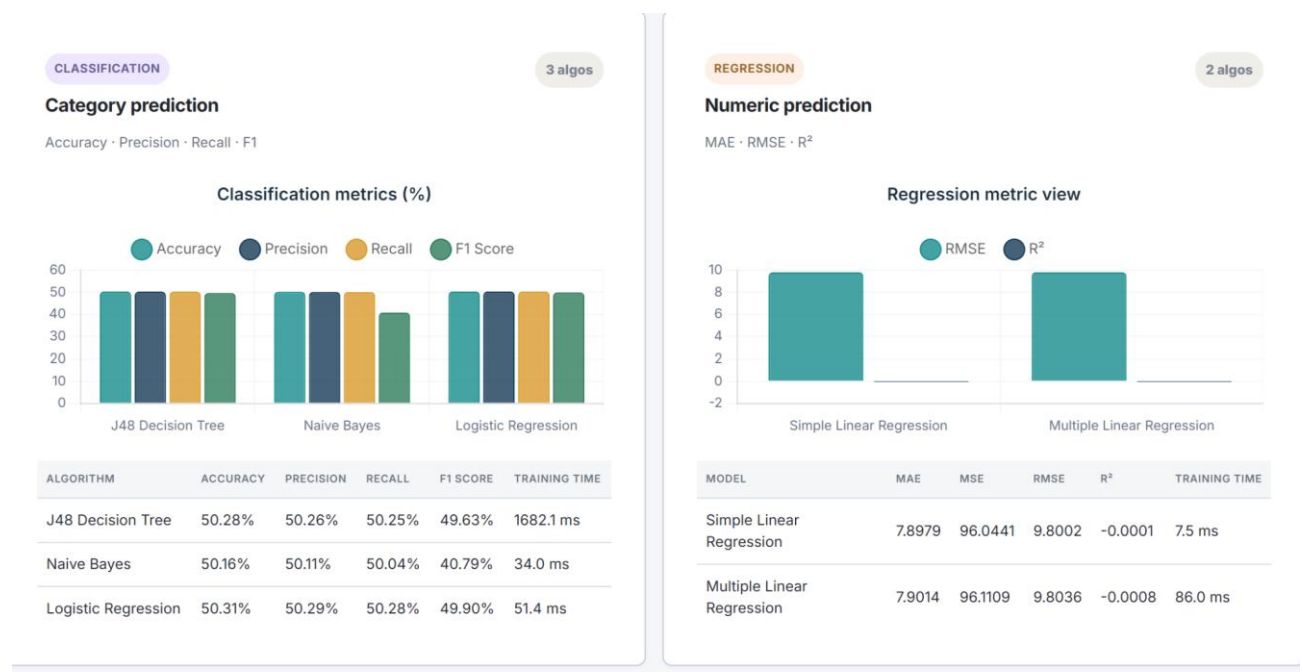


Figure 4: Supplied project screenshot showing classification and regression result panels together in the Results Hub.

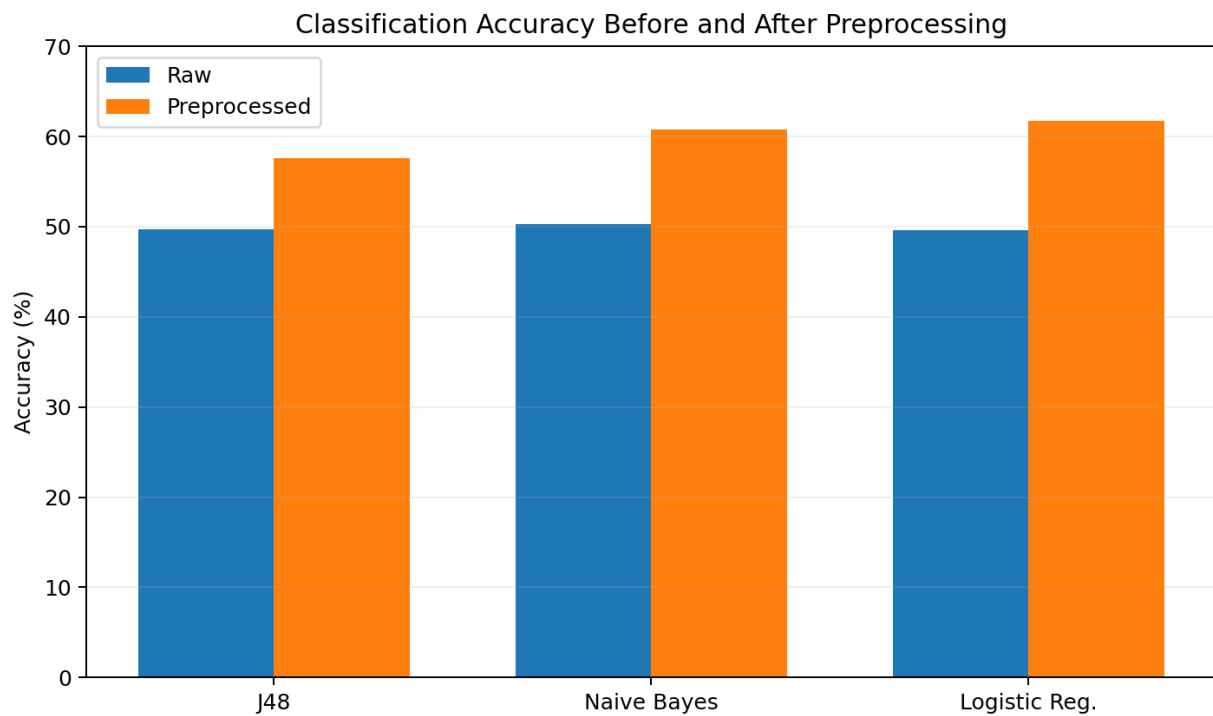
The screenshot is interface evidence from the supplied project state. Its displayed values should be treated as the captured UI result shown in the image. The verified source-run results reported below were generated separately from the bundled 50,000-row customer dataset and are intentionally kept distinct from the screenshot values.

9.2 Verified Classification Results

The verified classification experiment uses Customer_Churn as the target with an 80/20 stratified split and seed 42. The purpose of the experiment is not to force a high score but to observe the genuine effect of preprocessing on the three implemented classifiers.

Table 16: Verified Classification Results on the Bundled 50,000-Row Dataset

Algorithm	Raw Acc.	Raw Prec.	Raw Rec.	Raw F1	Prep. Acc.	Prep. Prec.	Prep. Rec.	Prep. F1
J48	49.69%	49.67%	49.69%	49.08%	57.56%	58.74%	57.56%	56.08%
Naive Bayes	50.25%	50.62%	50.25%	41.58%	60.80%	60.83%	60.80%	60.78%
Logistic Regression	49.63%	49.60%	49.63%	48.77%	61.77%	61.77%	61.77%	61.77%

**Figure 6: Verified classification accuracy before and after preprocessing.**

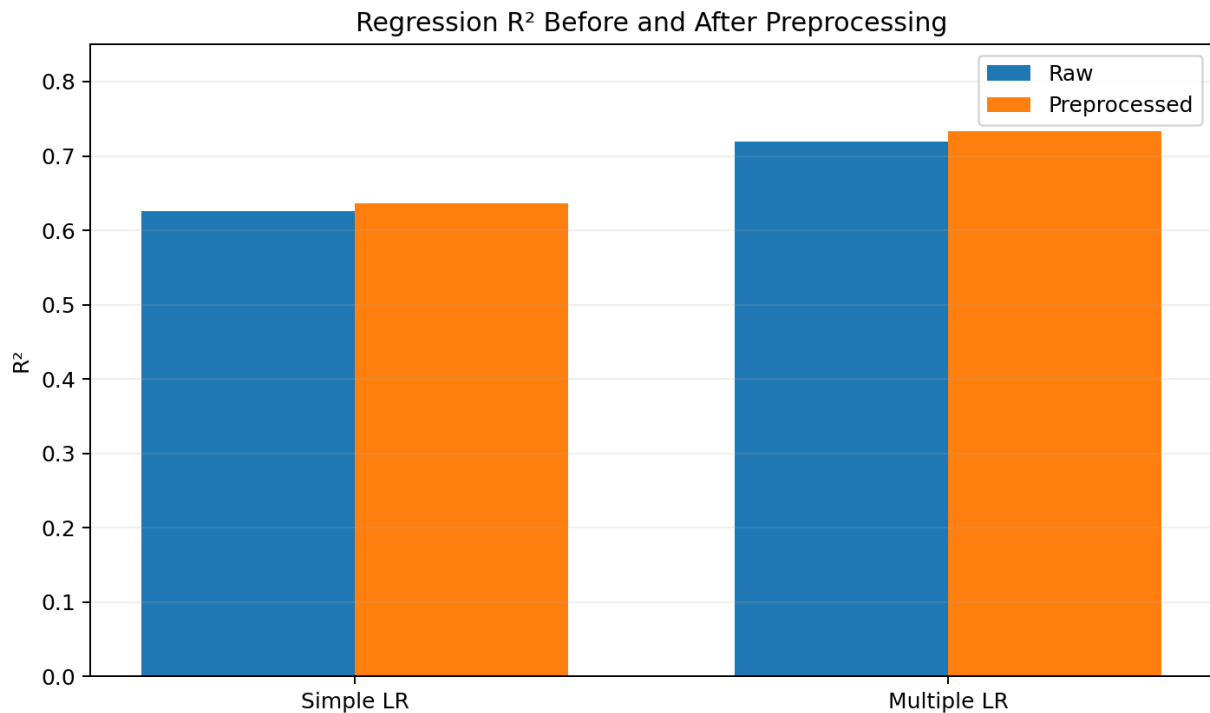
All three classifiers improve after the representative preprocessing pipeline. Logistic Regression reaches 61.77% accuracy in the verified preprocessed run, Naive Bayes reaches 60.80%, and J48 reaches 57.56%. The result demonstrates that data representation and missing-value treatment materially affect model behaviour; it should not be interpreted as a universal ranking of classifiers.

9.3 Verified Regression Results

The regression experiment uses Avg_Order_Value as the numerical target with an 80/20 split and seed 42. Simple Linear Regression uses Monthly_Income, while Multiple Linear Regression uses Monthly_Income, Discount_Usage_Pct, Products_Per_Order and Website_Visits.

Table 17: Verified Regression Results on the Bundled 50,000-Row Dataset

Model	Raw MAE	Raw RMSE	Raw R ²	Prep. MAE	Prep. RMSE	Prep. R ²
Simple Linear Regression	85.1596	106.8292	0.6254	84.2992	105.2404	0.6364
Multiple Linear Regression	73.4922	92.4363	0.7195	72.2162	90.1861	0.7330



Multiple Linear Regression remains the stronger of the two regression baselines in the verified run. Preprocessing reduces its MAE from 73.4922 to 72.2162, reduces RMSE from 92.4363 to 90.1861, and increases R² from 0.7195 to 0.7330. Simple Linear Regression also improves, but the additional predictors in the multiple model provide greater explanatory power on this dataset.

9.4 Association and Clustering Results

MineLab also contains the Apriori, FP-Growth and K-Means modules. The Market Basket dataset is intended for association analysis, where the principal outputs are frequent itemsets and rules ranked or filtered using support, confidence, lift and conviction. The K-Means module reports assignments, cluster sizes, centroids, silhouette score and WCSS. Because these tasks are unsupervised or pattern-discovery problems, they should be evaluated using their own metrics rather than compared directly with classification accuracy or regression R².

The clustering configuration can use K-Means++ style initialization, a fixed seed and a configurable number of iterations. The visualization module can display a two-dimensional projection, cluster-size summaries and elbow/WCSS information to make the discovered structure easier to inspect.

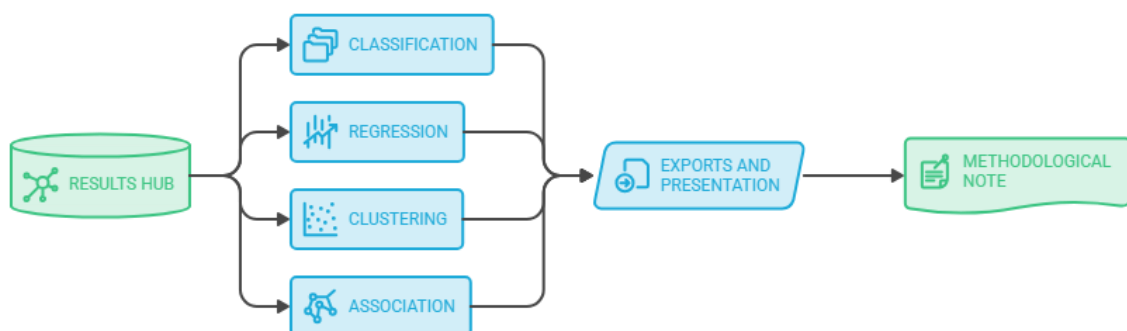
9.5 Task-Aware Integrated Comparison

The Results Hub brings the analytical families together without collapsing their metrics into a misleading universal score. The direct supervised-learning comparisons are especially useful because the classification models share a common target and held-out split, while the two regression models share the same target and evaluation protocol.

Table 18: Task-Aware Comparison Basis

Algorithm Family	Models	Primary Evaluation	Interpretation
Classification	J48, Naive Bayes, Logistic Regression	Accuracy, Precision, Recall, F1	Higher values indicate stronger held-out classification performance for the selected target
Regression	Simple LR, Multiple LR	MAE, MSE, RMSE, R^2	Lower error and higher R^2 are preferred, subject to target scale and context

MineLab Results Hub – Task-Specific Comparison



Made with Napkin

10. TESTING AND VALIDATION

10.1 Testing Strategy

Testing was performed at both the module and integrated-workflow levels. Individual data-processing and algorithm functions were exercised against bundled datasets, and the complete browser workflow was reviewed from dataset loading through preprocessing, mining, visualization and export. The supplied 50,000-row benchmark was also processed to verify that classification and regression metrics were generated from actual predictions.

Table 19: Functional Test Cases

Test ID	Scenario	Expected Result	Status
TC01	Open MineLab and select a bundled dataset	Dataset profile and navigation load correctly	Passed
TC02	Upload CSV / XLSX / XLS and switch datasets	Uploaded data becomes the active dataset	Passed
TC03	Run validation and attribute detection	Types, roles, missing values and statistics are displayed	Passed
TC04	Apply preprocessing options	Working dataset and quality statistics reflect selected operations	Passed
TC05	Run J48, Naive Bayes and Logistic Regression	Predictions and classification metrics are returned	Passed
TC06	Run Simple and Multiple Linear Regression	Predictions, equations and regression metrics are returned	Passed
TC07	Run K-Means	Cluster assignments, centroids and quality measures are returned	Passed
TC08	Run Apriori / FP-Growth	Frequent itemsets and association rules are generated	Passed
TC09	Open Results Hub, visualization and export	Tables, charts and export/report actions are available	Passed

10.2 Validation of Reported Experimental Results

The verified supervised-learning results were produced from the bundled 50,000-row customer dataset using Customer_Churn and Avg_Order_Value as the selected targets. Classification used an 80/20 stratified split with seed 42; regression used an 80/20 split with seed 42. The representative preprocessing run used median imputation, missing-value indicators and one-hot encoding, resulting in a $50,000 \times 54$ working representation. The resulting metrics were calculated from held-out predictions rather than entered manually.

11. LIMITATIONS AND FUTURE SCOPE

11.1 Limitations

The first limitation is persistence. MineLab relies on browser-side state and localStorage for selected settings and history, rather than a persistent enterprise data warehouse or multi-user data store. Restarting or clearing the browser session can therefore remove working state that has not been exported.

The second limitation is scale. The project is intended for interactive educational datasets. Algorithms such as Apriori can become computationally expensive when the number of unique items or maximum itemset size increases, while visualization of very large datasets may require sampling or reduction for responsiveness.

The third limitation is model realism. The implemented algorithms are compact educational versions designed to expose the underlying concepts. They do not provide the complete optimization, validation and production monitoring facilities of mature machine-learning platforms. Results therefore depend strongly on the dataset, feature representation and selected configuration.

Table 20: Limitations and Future Scope

Area	Current Limitation	Future Improvement
Persistence	Browser-side state and localStorage	Persistent warehouse, dataset metadata and model storage
Scale	Client-side in-memory processing	Sampling, optimized processing or distributed execution
Evaluation	Single held-out split in representative experiments	Cross-validation, confidence intervals and stronger validation workflows
Models	Educational implementations of standard algorithms	Advanced ensembles, tuned models and additional mining techniques
Visualization	Interactive charts designed for educational workloads	Richer diagnostics, larger-data rendering and automated report embedding

11.2 Future Scope

Future versions can add cross-validation and automated hyperparameter tuning, more advanced regression and classification models, stronger class-imbalance and target-quality warnings, model persistence, richer clustering diagnostics and expanded association-mining controls.

12. CONCLUSION

12.1 Conclusion

MineLab demonstrates how Data Warehousing and Data Mining concepts can be connected through one integrated academic analytics workspace. The application provides a common pipeline for dataset loading, validation, attribute detection, preprocessing, mining, evaluation, visualization and reporting, rather than treating each laboratory experiment as an isolated program.

The supervised-learning portion of the project contains three classification algorithms - J48 Decision Tree, Naive Bayes and Logistic Regression - together with two regression models - Simple Linear Regression and Multiple Linear Regression. The same preprocessing and evaluation infrastructure is reused across both branches. The verified 50,000-row benchmark demonstrates that preprocessing can materially change the feature representation and improve the observed classification and regression results.

The broader DWM workspace also includes K-Means clustering and Apriori / FP-Growth association rule mining. These modules use their own appropriate evaluation measures, reinforcing an important methodological point: different data mining tasks require different definitions of successful performance.

Overall, MineLab fulfils the goal of presenting a reusable, browser-based DWM environment that is suitable for laboratory work, practical examinations, demonstrations and self-learning. Its modular structure also provides a clear foundation for future extensions without changing the central analytical workflow.

12.2 DWM Assignment Mapping

Table 21: DWM Assignment Mapping

Assignment	Concept	MineLab Implementation
1	Data warehouse architecture	Conceptual DWM layers, ETL thinking and analytical data organization
2	WEKA / mining tools	Algorithm concepts demonstrated through custom JavaScript implementations; no direct WEKA service dependency
3	Dataset creation	Built-in synthetic datasets including preprocessing, Iris, Market Basket, Housing Prices and the 50,000-row benchmark
4	Data preprocessing	Missing values, duplicates, encoding, scaling, outliers and train/test splitting
5	Association rules	Apriori and FP-Growth with support, confidence, lift and conviction
6-7	Classification	J48 Decision Tree, Naive Bayes and Logistic Regression
8	Regression	Simple Linear Regression and Multiple Linear Regression
9	Clustering	K-Means with silhouette and WCSS evaluation
10	Mini project integration	Unified dataset analysis, Results Hub, visualization, reporting and export